All-in-one middleware for industrial human-robot-interaction

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Individual Mentoring Plan

Scope of the document

The first release of the IMP will describe the planning of the experiments for implementing the proposed HRI-enabled solutions, comprehensive of KPIs and Deliverables timeline. The IMP will be added to the SubGrant Agreement as an Annex before the first payment and will be updated after each milestone to ensure continuous monitoring and reporting.
The IMP constitutes the main tool to evaluate experiments' performance. More specifically, the proposals selected as an official beneficiary of the ARISE 12-month Support Programme will work to develop the proposed solution in collaboration with the ARISE consortium and will be evaluated in terms of performance in 4 stages (Figure 3):

- **Stage 1. Individual Mentoring Plan (1 month)**
- **Stage 2. Proof of Concept (4 months)**
- **Stage 3. Deployment and Experimentation (5 months)**
- **Stage 4. Launch sharable HRI modules (2 months)**

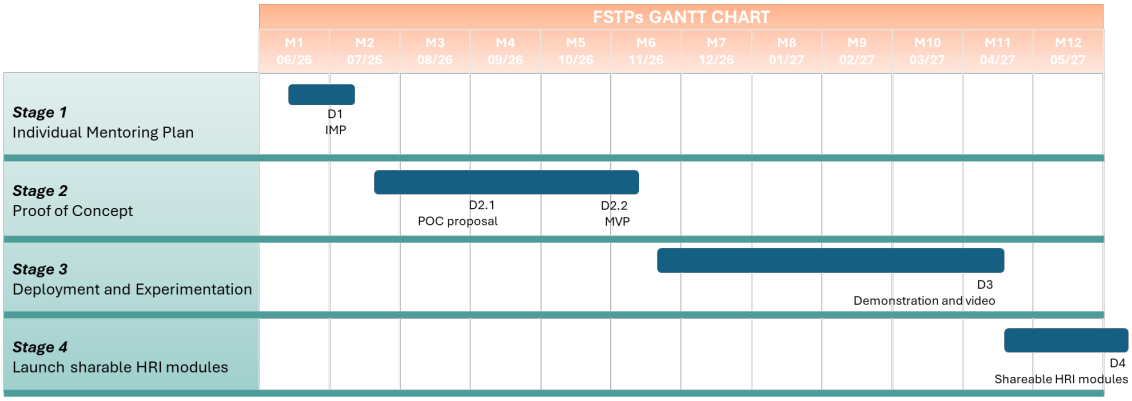


Figure 3: GANTT of the ARISE 12-month Support Programme.

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Before starting

In each section, the corresponding instructions in italics might help to complete them. In some cases, open text is required to describe the activities and expected outcomes, in other cases tables are requested to be filled.

Sections coming from the application form are in blue color:

- copy and paste the text written for the proposal or leave them empty and attach the full proposal as an annex;
- use these sections to integrate any possible feedback received during the evaluation and selection process.

Please, use Exo 2 font with a size dimension of 10.

To describe a code, please follow the example in the following:

```
$ ls -l
```

Please consider that the information collected in the present Individual Mentoring Plan could be used on the ARISE website and social media to present the Open Call winner. In case some information is strictly reserved, please report it in the specific sections. The completion of this document constitutes the acceptance to extract information for the publication.

PROJECT IDENTIFICATION	
Project title	**Posture-Aware Cobot Assistance in Packaging**
Project Acronym	**POSTURE**
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0. Recommendations

The following table collects the recommendations provided as starting point from your mentor (referred to as Stage 0) and those that emerge at each milestone review (Stage 1-4). Recommendations are differentiated based on the category (Technical, Ethical, Business, etc.) and on the priority level (Critical, Preferred, Optional). The beneficiary is required to address each recommendation outlining the actions taken and indicating the relevant sections in the document.

Stage	Recommendations	Response and actions taken

1. Project short description

POSTURE is an ergonomics-first rework and packaging cell software and hardware solution co-created by MMVR and the University of Alicante (UA), and validated at end-user site Pepri Toys, a toy manufacturer based in Ibi, Alicante. The core idea is simple: rather than requiring operators to adapt their bodies to the workstation, the workstation adapts to each operator. An adaptive UR10 collaborative robot and a height/tilt-adjustable fixture automatically present and hold Blow molded parts at each operator's neutral posture position, eliminating the strain-intensive gripping and rotation that currently drives musculoskeletal risk on the line. Operators retain full control via a touch-screen interface with optional voice commands, and can pause, nudge, or resume at any point without losing process context.

The solution is built on open standards: ROS 2 (Vulcanexus) for robot orchestration and FIWARE (Orion-LD with NGSI-LD) for real-time analytics. A reusable ROS 2 skill called ErgoAssist encapsulates all posture-aware behaviors and will be contributed to the ARISE open catalog. The FIWARE-powered dashboard logs ergonomics proxies, safety events, cycle time, and first-pass yield in real time, with automated alerts and reports.

The project targets one level of improvement in RULA/REBA ergonomic risk scores, a 15% reduction in perceived workload (NASA-TLX), and a 5–10% throughput increase. A working proof of concept is planned for Month 4 and a TRL 7 demonstrator for Month 10, leaving two months for exploitation and dissemination.

1.1 Selected challenge

- ☐ *CH1: Dismantling and assembly of high-value products*
- ☐ *CH2: Complex Product Picking in industrial warehouses*
- ☐ *CH3: Flexible Collaborative robots*
- ☐ *CH4: Smart Programming*
- ☐ *CH5: Enhanced robot functionality through multimodal HRI interactions*
- ☐ *CH6: Fetch & carry tasks in healthcare environments*
- ☐ *CH7: HRI for improving the efficiency of workers in high precision flexible tasks*
- ☐ *CH8: HRI for improving ergonomics in high precision tasks*

1.2 Objectives and Ambition

POSTURE validates an Industry 5.0 scenario in which the robot reduces physical strain for manual operators at a factory. The collaborative robot yields to human intent. Operators can make micro-adjustments via a clear HMI and optional voice commands, keeping them in control at all times.

POSTURE implements posture-aware HRI with the fixture (fixture in this context is the post-processing working table) which auto-sets height from a per-operator profile. Additionally, the robot adjusts the blow molded part with personalizing tiling to the operator. On the analytical side, a FIWARE-powered condition monitoring application provides real-time

dashboards, a cycle and ergonomics historic log, and automated alerts for posture-risk spikes per SKU or shift.

The advanced all-in-one middleware use cases include: replaying ROS 2 logs into a lightweight digital twin to A/B test fixture angles and speeds before deployment; optional on-premise inference that estimates posture-risk proxies from non-identifying pose signals and exposes them to FIWARE via NGSI-LD; and heterogeneous IT connectors so production and quality events flow to both a modern REST ERP and a legacy CSV-based manufacturing execution system.

The reusable ROS 2 ErgoAssist skill exposes standard actions: **PresentAtAngle, HoldAndRotate, ConfirmQuality, YieldToHuman,** with per-operator and per-SKU parameters, and will be submitted to the ARISE catalog with full tests, tutorials, and example configurations.

1.3 Human centricity

Before any build activity, the team conducts on-floor observations and structured interviews and cell leads across all shifts to map real pain points:. These insights directly become requirements.

Before initiating any hardware or software engineering activity, the project consortium will execute an exhaustive, on-site industrial diagnostic workflow at the Pepri Toys production plant. Grounded in Lean Management methodologies and qualitative audits will be conducted with line managers and operations **(at least 8 Pepri Toys operators)** leads to analyze the end-to-end manufacturing flow.

This systemic approach aims to isolate and rectify upstream bottlenecks and operational variances prior to automation, preventing the digital encapsulation of waste (muda) and ensuring that the human-robot collaboration cell operates on an inherently optimized process. Simultaneously, operators assigned to the target packaging cell will undergo semi-structured interviews and participatory ergonomics mapping to identify localized physical and cognitive pain points, such as wrist and neck strain during holds and rotations, visibility at awkward angles, time pressure, and interruptions, sustained muscular load, musculoskeletal vulnerabilities during high-frequency cycles, and adverse sightline constraints.

To guarantee high-fidelity data acquisition while enforceably minimizing floor disruptions, down-time, and repetitive physical interventions on the production line, the team will implement a non-intrusive spatial digitization protocol. Utilizing the advanced Inside-Out tracking and high-resolution LiDAR-assisted pass-through sensors of the Meta Quest 3 headset, the team will execute real-time 3D Scene Mesh generation and spatial anchors mapping of the work cell. This dense point-cloud data and spatial geometry, complemented by 360-degree stereoscopic video captures, will be synthesized to reconstruct a high-precision digital environment. This assets pipeline will directly feed the system's lightweight Digital Twin, providing an immutable, dimensionally accurate foundation for offline virtual prototype validation, A/B simulation of collaborative fixtures, and post-visit ergonomics assessment without impacting the plant's operational throughput.

The synthesis of this Lean diagnostic, qualitative operator mapping, and spatial digitization will culminate in a formal User Needs Document (UND). This document will bridge human-centric feedback and raw spatial data into explicit, traceable technical requirements, functional specifications, and safety constraints that will directly govern the subsequent software architecture, ROS 2 interface contracts, and mechanical design phases.

Usability is a release gate and must reach 75 or above overall, and if any subgroup scores below that threshold, a redesign sprint is triggered before the project advances.

The HMI architecture is built upon an inclusive and highly adaptive design paradigm to accommodate diverse operator profiles. The user interface features high-contrast, glove-friendly interactive touch targets to prevent misinputs under industrial conditions, dynamic left/right-handed mirroring to support ergonomic workspace layouts, and adjustable typography scaling for varying visual acuities. Standardized, color-blind-safe iconography is implemented alongside high-visibility color palettes to guarantee readability.

All visual labeling, contextual system alerts, and XR training micro-lessons are fully localized and dynamically switchable between Spanish, Valencian, and English, ensuring cognitive accessibility and minimizing training barriers across a multicultural workforce

Operator recruitment for validation must ensure we include:

- Gender balance (representation of women and men)

- Age distribution (ranging from 20 to 55+ years old) to account for varying physical stamina

- Multi-tier experience levels (from novice trainees to expert, highly experienced operators)

- Handedness diversity (including both right- and left-handed workers)

- Anthropometric diversity, explicitly recruiting a wide range of operator heights to validate the adaptive workspace's mechanical range.

Test results are disaggregated by subgroup, and if any group underperforms, targeted UI or training adjustments are made and retested before the Month 10 TRL 7 demonstration.

Privacy is integrated by design across all system layers. The computer-vision pipeline utilizes non-intrusive posture-tracking algorithms where raw video streams of the operators are processed strictly in transient memory at the edge; no raw video footage, RGB images, or biometric facial identifiers are ever saved or stored on local or cloud storage. Instead, the system instantly discards the visual data after extracting anonymous, mathematical 2D/3D skeletal joint coordinates (skeleton wireframes). These non-identifying skeletal figures are the sole telemetry used to calculate real-time RULA/REBA joint angles and ergonomic risk parameters.

Furthermore, the data architecture enforces explicit informed consent, pseudonymization of all operational logs, strict role-based access control (RBAC), and a hard 90-day retention limit on raw telemetry. Operators retain the right to opt out of downstream analytics at any time without penalty, retaining full access to standard cell operation. A cross-shift Operator Advisory Group, including health and safety representatives, co-owns HMI and workflow changes and signs off on releases. A one-page, plain-language summary of data processing and purpose limitation is physically posted at the work cell, alongside a direct, transparent grievance and feedback channel.

1.4 Industrial added value through human-robot interaction

Across the EU there are approximately 70,000–100,000 manual packaging and rework stations. Assuming 30–40% are high-precision, repetitive, and ergonomically exposed, the total addressable market for an ergonomics-first cobot cell is 20,000–40,000 stations. The three-year serviceable addressable market is 15,000–25,000 stations, of which 5,000–8,000 already operate or plan cobot deployments. The serviceable obtainable market within 18–24 months, reachable via Pepri's network and partner integrators, is 300–600 stations.

ErgoAssist transfers strain-heavy holds and rotations to the robot and standardizes low-effort micro-adjustments, delivering at least one level of RULA/REBA improvement, 50% fewer near-misses, 15% lower NASA-TLX workload, and 5–10% higher throughput. With 3,000–4,500 productive hours per year at €50–€100 value-add per hour, this yields €75,000–€450,000 annual benefit per station against a deployment cost of €40,000–€85,000, enabling 6–12 month payback and 60–250% Year 1 ROI.

Replicability is built on three layers: the ErgoAssist ROS 2 skill with operator profiles, SKU presets, and fixture abstraction for multi-plant reuse; the FIWARE application with portable NGSI-LD models for production, quality, ergonomics, and safety events; and thin REST/CSV ERP-MES adapters that do not affect safety-critical control. The open reference implementation, clear ROI figures, and an HRI inclusivity checklist together shorten procurement cycles and de-risk trials for SMEs.

1.5 Technical and Business KPIs

Objective	KPIs			
	Description	Type	M1 value	M12 value
General business objective	*Name of the KPI*	*Technical, Business*	*Initial value*	*Targeted value*
Reduce ergonomic risk	RULA/REBA score on targeted subtasks	Technical (Ergonomics)	~5–6 (medium-high risk)	≤4 (at least 1-level reduction)
Reduce workload	NASA-TLX overall score	Technical (Human Factors)	~55–65 (estimated for repetitive precision assembly)	≤47 (–15%)
Increase Throughput*	Cycles per hour	Business	This will be measured on our site visit.	+10%
Operator acceptance	% operators preferring the new cell	Business	0% (pre-deployment)	≥80%
User friendly interface	% operators identifying user friendliness	Technical (Usability UX)	0% (pre-deployment)	≥80%

1.6 Current scenario vs future scenario

Current situation	Short-term scenario (at the end of the project)	Long-term scenario (in the following years)
Operators at Pepri manually hold, rotate, and position the different pieces during rework and packaging, generating sustained wrist, neck, and shoulder strain. There are complaints. No ergonomic data is systematically collected.	A validated TRL 7 ergonomics cell is operational at one Pepri station, covering 3 SKUs and tested with 10+ operators. ErgoAssist is published in the ARISE catalog. Measurable gains in RULA/REBA, NASA-TLX, and throughput are documented with rigor.	The ErgoAssist skill and FIWARE bundle are deployed across multiple Pepri stations and licensed to external integrators. The solution is replicated at 40–55 stations across 18–25 sites within 3 years, generating recurring support revenue and contributing to European adoption of human-centric industrial HRI.

1.7 State of the Art of the technology

Most existing collaborative robot deployments in packaging and assembly focus on cycle time optimization or simple pick-and-place operations. Very few solutions offer posture-aware primitives that adapt dynamically to individual operator profiles, human-in-the-loop exception flows based on pause–nudge–resume rather than full E-stop and reteach cycles, or built-in ergonomics analytics connected to enterprise systems via open standards.

ErgoAssist goes beyond the current state of the art by combining per-operator preset management, force-limited smooth rotation primitives that eliminate biomechanically harmful grip patterns, and a real-time FIWARE-powered monitoring layer that makes ergonomics data auditable and portable. The use of ROS 2 (Vulcanexus) and NGSI-LD as open de-facto standards ensures interoperability with other HRI components and enables replication without proprietary middleware dependency. The digital twin capability for A/B testing fixture parameters before live deployment is also novel at this scale and TRL for SME-facing solutions.

1.8 Project team

Pepri Toys, MMVR, and the University of Alicante (UA) form a highly robust, cross-functional consortium characterized by proven leadership, technical execution capability, and direct commercial market access. The team actively incorporates a strong gender balance across critical coordination and leadership roles.

At MMVR,

- Beatriz Zambrano (Project Coordinator): Manages overall consortium governance, cross-partner communication, milestone deliverables, agile sprint planning, and the business development pipeline to guarantee seamless time-to-market execution.
- Jose Garcia (Lead Full-Stack & Integration Engineer): Responsible for core HMI engineering, ROS 2 integration layers, and WebXR/Unity deployment for the interactive training modules.
- Jesus Silva (Senior UX/UI Designer): Leads user-centered design paradigms, cognitive ergonomics, accessibility frameworks, and inclusive HRI validation.

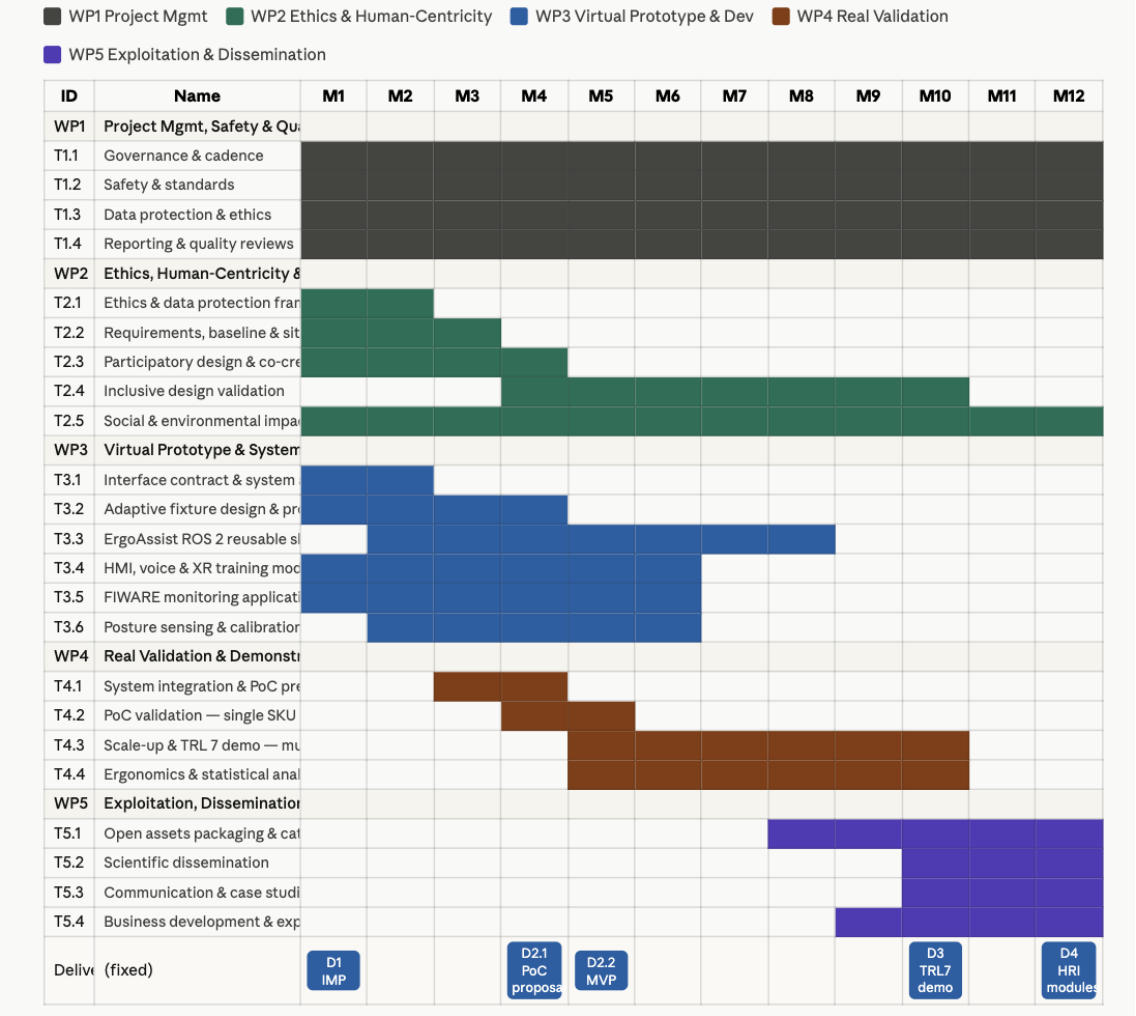
At Pepri,

- Roberto Martínez (Executive Sponsor): Allocates plant resources, unblocks operational dependencies, and steers the internal adoption strategy.
- Carolina Prieto (Site and Operator Lead): Coordinates the Operator Advisory Group, inclusive sampling recruitment, and shift-by-shift validation schedules.
- Pedro Pareja (Safety and Quality Officer): Oversees compliance protocols for manufacturing and safety standards, and QA gating.
- Raúl Teruel (Production Lead): Anchors manufacturing deployment and shop-floor integration on the live production line.

At University of Alicante,

- Dr. José García (internal coordinator): Manages the university's resource allocation, technical alignment with the ARISE middleware ecosystem, and oversees the implementation of AI/robotics primitives in the cell.
- Dr. David Mullero (Lead Ergonomics Researcher): Coordinates the posture-sensing computer vision pipeline and ensures its calibration with standard industrial RULA/REBA ergonomics protocols.
- Dr. David Ortiz (Senior Data Analyst): Handles the mathematical data architecture, evaluating the psychometric NASA-TLX results and ensuring all validation reporting is auditable.
- Michael Fernandez (PhD Researcher & Industrial Engineer): Technical Lead Developer. Drives the physical software development, ROS 2 ErgoAssist skill architecture, and CI/CD catalog packaging, applying Lean Manufacturing principles to align the cell with the shop-floor layout

2. Implementation plan



2.1 Detailed description of tasks

WP *1 - Project Management, Safety and Quality*	Duration: M1–M12
Objectives	*Ensure coordinated execution across all partners; manage scope, schedule, budget, risks, ethics compliance, GDPR, safety standards, and quality gates throughout the full project lifecycle.*
Tasks	**T1.1 – Governance and Cadence (M1–M12)** Bi-weekly technical sprints, monthly steering committee meetings, and continuous RAID and RACI management. Output: Project Handbook (M1) **T2 –Safety and Standards (M1–M12):**

	Compliance engineering for ISO 12100, ISO 10218-1/-2, and ISO/TS 15066; collaborative safety validation; safety event logging to FIWARE. Output: Safety Plan (M1); **T1.3 – Data Protection and Ethics (M1–M12)**: DPIA, operator consent procedures, pseudonymization. Output: DPIA and Consent Framework (M1); **T1.4 – Reporting (M1–M12):** Quality assurance, PoC gate review, and TRL 7 gate review. Output: PoC Gate (M4); TRL 7 Gate (M10).

WP *2 -Ethics, Human-Centricity and Inclusivity*	Duration: M1–M12
Objectives	*Embed ethics-by-design and human-centred design principles across all project phases; ensure GDPR compliance, informed consent, and operator wellbeing; establish confirmed baselines and site readiness; guarantee inclusive access for diverse worker profiles; and monitor social and environmental impact*
Tasks	**T2.1 – Ethics and Data Protection Framework (M1–M2)** Develop the DPIA, operator consent forms, pseudonymisation procedures. Map the project against the EU relevant standards (ISO 9241-210, ISO 11226 etc.) Output: DPIA and Consent Framework (M2). **T2.2 –Requirements, Baseline and Site Readiness (M1–M3)** Operator interviews and on-floor observations at the PepriToys station, anchored to the site visit. Formal KPI and baseline specification for direct measurement. Station hardware confirmation, IT/OT network preparation, and FIWARE component deployment. Output: Requirements and KPI Specification (M2); **T2.3 – Participatory Design and Operator Co-Creation (M1–M4** Establish the cross-shift Operator Advisory Group including health and safety representatives. Run paper mockup tests with at least 6 operators before any code is written. Conduct think-aloud tests in realistic factory conditions: noise, time pressure, gloves. Post a plain-language data notice at the cell and provide a grievance channel. Output: Validated wireframes and HMI requirements (M2); **T2.4 – Inclusive Design Validation (M4–M10):** Iterative SUS and HRI-trust scale testing disaggregated by gender, age, handedness, and experience level. Redesign sprint triggered if any subgroup scores below SUS 75. Output: UX Validation Reports (M6, M10) **T2.5 – Social and Environmental Impact Monitoring (M1–M12):** Pre- and post-deployment survey tracking job security perception (target: –25% reporting threat) and job quality (target: +30% reporting improvement). Track scrap rate, energy per part, and idle time in FIWARE as environmental KPIs. Document the avoided embodied footprint of the retrofit-first approach. Output: Impact Assessment Reports (M5, M10).

WP *3 - Virtual Prototype and System Development*	Duration: M1–M8
Objectives	*Develop and validate all technical components of the POSTURE solution in a controlled environment before live deployment, including the adaptive fixture, the ErgoAssist ROS 2 reusable skill, the HMI and XR training modules, and the FIWARE monitoring application.*
Tasks	**T3.1 –Interface Contract and System Architecture (M1–M2)** Define and formally deliver the full ROS 2 interface specification, message formats, action and service definitions, robot state outputs (READY, MOVING, YIELDED, EXCEPTION, CYCLE_COMPLETE), response time requirements, and error handling behaviour. This is a critical dependency for offline integration testing. Output: ErgoAssist simulator (M2). **T3.2 –Adaptive Fixture Design and Prototype (M1–M4):** Mechanical design and fabrication of the height- and tilt-adjustable fixture (post-processing working table) with safety locks, PPE compatibility, and operator preset support. Output: Prototype fixture and bill of materials (M3);Operator preset validation report (M4). **T3.3 – ErgoAssist ROS 2 Reusable Skill (M2–M8):** Implement posture-aware collaborative robot skills including force-limited motion, per-operator and per-SKU preset management, and autonomous exception recovery (target: ≥85%). Package with CI tests, tutorials, and example configurations for submission to the ARISE catalog. Output: Proof of Concept PoC delivery (M4) and update (M8) **T3.4 – HMI, Voice Interface and XR Training Modules (M1–M6):** Build the inclusive touch-screen HMI with user friendly targets, operator profile management, and optional voice commands (stop, rotate, continue, help), validated at ≥80% recognition accuracy under factory noise. Develop five XR training micro-lessons in Unity covering robot introduction, profile creation, first safe shift simulation, exception handling, and ergonomics self-care. Languages: Spanish and Valencian primary, English secondary. Target: first safe operation in ≤30 minutes after training. Output: Complete training module suite (M6). **T3.5 – FIWARE Monitoring Application (M1–M6):** Define NGSI-LD data models for production, quality, ergonomics proxies, safety events, and energy consumption. Implement real-time dashboards, a cycle and ergonomics historian, automated posture-risk alerts, and weekly PDF/CSV report generation. Output: Live dashboard (M4); and updated with alert rules and reports (M6). **T3.6 – Posture Sensing and Calibration (M2–M6):** Calibrate camera and sensor posture proxy pipeline against reference RULA/REBA assessments. Validate operator presets across PPE, shift lighting, and height variability.

	Output: Calibration report (M6)

WP4 –Real Validation and Demonstration | Duration:	M3–M10
Objectives	*Validate the integrated POSTURE solution against all technical and human-outcome KPIs in the real production environment at Pepri Toys, progressing from a single-SKU proof of concept to a multi-SKU, multi-operator TRL 7 demonstrator.*
Tasks	**T4.1 – Proof of Concept Validation — Single SKU (M3–M5):** Validate the complete system on one SKU within the 1 shift with approx 25 cycles in each shift. Acceptance criteria: ≥1-level RULA/REBA improvement on targeted subtasks; SUS ≥75 with ≥8 operators; zero unsafe contacts in predefined exception scenarios; Output: PoC Demonstrator and Validation Report (M3) and updated (M5) **T4.2 – Scale-Up and TRL 7 Demonstration. (M5–M10):** Validate the complete system on one SKU within the 3 different shifts, with approx 50 cycles in each shift, with ≥1-level RULA/REBA reduction; 15% NASA-TLX workload; +5–10% throughput; SUS ≥75 with no subgroup below threshold. Output: TRL 7 Demonstrator, final update and Full Validation Report (M10). **T4.3 – Ergonomics and Analysis (M5–M10):** Design and execute all evaluation protocols. Disaggregate results by gender, age, and handedness. Run monthly reviews feeding near-miss root causes and micro-break adherence back into design sprints. Output: Statistical Analysis Reports (M10).

WP5 – Exploitation, Dissemination and Open Assets |	M8–M12
Objectives	*Maximise the impact and replicability of POSTURE results through open asset publication, scientific dissemination, business development, and a structured exploitation plan for commercialisation beyond the ARISE project.*
Tasks	**T5.1 – Open Assets Packaging and Catalog Submission (M8–M12):** Finalise the ErgoAssist ROS 2 skill with CI tests, datasets, and a compatibility matrix for major cobot models and fixture types. Publish NGSI-LD data models for ergonomics, quality, and safety events. Release the connector SDK (REST/CSV) and an HRI Inclusivity and Ergonomics checklist with training micro-lessons. Output: Catalog-ready ErgoAssist module (M8); full open asset package (M12). **T5.2 – Scientific Dissemination (M10–M12)**: Prepare and submit two open-access publications. Participate in at least 2 industrial HRI or robotics conferences. Output: 2 submitted manuscripts (M12); ≥2 conference presentations.

T5.3 – Communication and Case Studies (M10–M12): Produce two case studies (English and Spanish) with pre/post ergonomics and safety data from Pepri Toys. Publish project results on the ARISE website and MMVR social media at each milestone.
Output: 2 case studies (M12)

T5.4 – Business Development and Exploitation Plan (M9–M12):
Finalise the exploitation plan covering revenue streams, go-to-market strategy, and 3-year projections. Certify two partner integrators to enable scale without headcount growth. Begin active sales outreach targeting 5–10% of new ergonomics cobot retrofits in the initial region within 3 years.
Output: Exploitation Plan (M12); ≥2 integrator certifications in progress.

2.2 Risk evaluation and mitigation plan

Barrier/Challenge (P = [1-5]; I = [1-5])	Associated risk	Action to mitigate the risk
Interface contract delay (P: 4; I: 5)	Metamedics (Jose Garcia) cannot begin software development until the ROS 2 interface specification, message formats, robot states response times, and error behaviours, is delivered by University of Alicante (Michael). If delayed beyond M1, the Month 4 PoC gate is directly at risk.	Interface contract is a formal deliverable due at the end of M1 (T3.1). An ErgoAssist stub/simulator that responds to all defined messages with simulated outputs is delivered by UA to MMVR by M2, enabling offline integration testing without physical presence at Pepri Toys. Progress tracked at every bi-weekly sprint from M1.
Station hardware definition delayed (P: 3; I: 3)	If Pepri does not confirm the defined SKU, operators, HMI target hardware (industrial touch panel, tablet, or panel PC; operating system) by M2, technology providers cannot commit to the final technology stack and risks building for the wrong platform.	MMVR builds an HMI version for internal testing from M1, independent of hardware decision. Hardware specification is a formal output of T2.2 site readiness (M3). If not confirmed by M2, escalated to monthly steering committee for decision.
ROS 2 to FIWARE latency exceeding 2-second threshold (P: 2; I: 4)	In the real factory network environment, OT→IT pipeline latency may consistently exceed the 2-second dashboard KPI , degrading real-time alert functionality and missing a	Latency measured from first integrated dry-run (M3) and tracked as a primary KPI. Network configuration validated in T2.2. If above threshold at M3, UA optimises the Orion-LD bridge before the M4 PoC gate. Edge buffering available as fallback.

	primary system target.	
Operator recruitment and scheduling constraints (P: 3; I: 3)	*Pepri production pressures may limit the number of operators available for testing sessions across all shifts,*	Operator Advisory Group established in M1 with HR and health and safety involvement. Carolina Prieto (Site Lead) formally responsible for scheduling. Test sessions designed to be ≤45 minutes and runnable during shift transitions to minimise production impact.
Baseline data gaps after site visit (P: 2; I: 3)	*If the July 2025 site visit cannot capture sufficient baseline measurements PoC delta calculations will lack a credible reference point.*	*An early fallback visit will be scheduled in M1/June to secure a dual-visit baseline acquisition window. Additionally, the team will use Meta Quest 3 LiDAR scene meshing and 360° video to reconstruct the work cell as a Digital Twin, enabling highly precise, remote offline measurements and completely preventing the need for disruptive follow-up physical visits.*

3. ARISE support services

ARISE partners will support the open call beneficiaries through a series of services. Each service will focus on specific topics and be tailored to the specific needs of the experiments. This section provides a brief description of the services and the expected output. Open Call beneficiaries will be required to provide preliminary input useful for ARISE partners to start the activities.

3.1 FSTP Support Service I: Standardization and interoperability using the ARISE middleware

The **main objective** of this support service is to ensure a suite of interoperable tools and components that enable HRI programming, testing, and deployments in industrial settings. The software development lifecycle of a module using Industrial HRI, in fact, involves several stages and, throughout the entire lifecycle, the adherence to standards and conventions is important to ensure that the module is interoperable with other HRI interaction components and tools. The experts will provide this service which builds on the model-based approach of the EU Digital Industrial Platform for Robotics and relies on ROS4HRI, ROS 2, and FIWARE's NGSI-LD technologies, all of them open de-facto standards to ensure a suite of interoperable tools and components.

The **expected outcomes** at each review milestone are the following:

- Stage 1. Individual Mentoring Plan (1 month - M01): Establish the framework for support, including initial requirements and tailored mentoring plans for integration ARISE middleware components.
- Stage 2. Proof of Concept (4 months - M02-M05): Demonstrate interoperability and adherence to standards via a working prototype showcasing the integration of HRI modules within ARISE framework.

- Stage 3. Deployment and Experimentation (5 months - M06-M10): Full deployment of middleware-supported modules, with real-world experimentation ensuring scalability and robustness.
- Stage 4. Launch shareable HRI modules (2 months -M11-M12): Finalize the development of shareable HRI modules compliant with ARISE middleware, publish them with supporting documentation, and prepare them for widespread adoption.

According to this, the service will follow a structured and collaborative approach to address the specific needs of the open call experiments:

- *Review*: Evaluate the capabilities of ROS4HRI, ROS2, and FIWARE's NGSI-LD technologies to guide the development and standardization of interoperable tools and components. This analysis will also consider specific hardware, IoT technologies, and data models.
- *Documentation Support*: Provide detailed, standard-level documentation (e.g. messages, topics, service specifications, and API references) to facilitate technology provider integration.
- *Targeted Support*: Develop tutorials and examples to demonstrate standardized practices and solutions (preferred reusable approach). Offer one-to-one remote mentoring sessions tailored to address individual project challenges.

3.2 FSTP Support Service II: Open-source tools and off-the-shelf modules for Industrial HRI

The **main objective** of this support service is to help technology providers to integrate ROS4HRI into their solutions from the design and implementation stages. The task includes support at the architecture design stage, looking into how existing or new solutions can be modified to be ROS4HRI compliant, advising on existing technical solutions (e.g. open-source software packages), and providing technical support to implement and integrate the Industrial HRI modules.

The **expected outcomes** at each review milestone are the following:

- Stage 1. Individual Mentoring Plan (1 month - M01): Develop a tailored support plan detailing specific tools, modules, and steps required for ROS4HRI integration.
- Stage 2. Proof of Concept (4 months - M02-M05): Validate the integration of open-source tools and off-the-shelf modules within the provider's solution architecture.
- Stage 3. Deployment and Experimentation (5 months - M06-M10): Ensure the deployment of ROS4HRI-compliant solutions in operational environments, focusing on usability and effectiveness.
- Stage 4. Launch sharable HRI modules (2 months -M11-M12): Publish the final modules as open-source tools with accompanying documentation and examples, fostering a community of practice and adoption.

According to this, the service will follow the key activities to address the needs of the experiments:

- *Guidelines*: Provide guidance to ensure ROS4HRI integration from initial architecture design to final implementation.
- *Ad-hoc Support Requests*: Address specific issues raised by open call beneficiaries through tailored technical advice.

- *Documentation Support*: Create and share tutorials, examples, and best practices to guide providers in integrating open-source tools and modules (reusable approach preferred).
- *Hands-On Support*: one-to-one remote sessions to resolve implementation challenges and provide direct, hands-on assistance for integrating ROS4HRI modules into industrial applications.

3.3 FSTP Support Service III: Human-centricity by design using the ARISE SSH framework

POSTURE contributes to two ARISE core priorities.
On inclusion, the solution adapts the workstation to the worker rather than the reverse. Operator recruitment includes females and and includes workers over 50, novice and experienced operators, and all shifts including night shifts.
Results are disaggregated by subgroup, and targeted redesign is triggered if any group underperforms.
Capacity gains from reduced injuries and interruptions are explicitly redeployed to quality improvement activities rather than headcount reduction.
A pre- and post-deployment survey tracks whether operators feel the robot threatens job security (target: −25%) and whether they report improved job quality through safer, less tiring, and more predictable work (target: +30%).

On the environment, the retrofit-first approach avoids the embodied footprint of a new automation cell, estimated at 1–3 tonnes of CO_2e per station based on the bill of materials that would otherwise be procured. Scrap reduction of 5–10% on rework steps and energy consumption reduction of 5–8% per good part through smoother assisted rotations and automatic standby are tracked directly in FIWARE using NGSI-LD entities, ensuring improvements are auditable and reproducible across sites.
Open assets, ErgoAssist, NGSI-LD models, an HRI Inclusivity and Ergonomics checklist, and training micro-lessons, lower adoption barriers for SMEs across Europe

Ethical and legal framework

Regulatory sources	Relevant content	Legal/ethical issues
GDPR — Regulation 2016/679	Personal data processing, consent, data minimisation, purpose limitation, retention limits	Operator profile data and posture proxy logs are personal data; DPIA required; pseudonymisation and 90-day retention limit implemented from M1
ISO 12100 / ISO 10218-1/-2 / ISO/TS 15066	Collaborative robot safety, risk assessment, speed and force limits, stopping distances	Full compliance required for PoC and TRL 7 safety sign-off; all safety events logged to FIWARE
ISO 9241-210 / ISO 9241-11	Human-centred design process and usability	SUS ≥75 applied as release

	measurement	gate at every validation milestone; redesign sprint triggered if any subgroup falls below threshold
ISO 11226 / EN 1005	*Working postures and manual force limits*	Reference standard for RULA/REBA baseline methodology, target-setting, and ergonomics assessment protocols

Anticipated ethical concerns	Description	Planned actions to mitigate
Use of posture and movement data	Camera and sensor data used to estimate ergonomic risk proxies during operation	Only derived non-identifying metrics (e.g. time in risk band, rotation duration) reach the analytics layer; no biometric identifiers stored; explicit informed consent obtained before any data collection begins
Operator perception of surveillance	Continuous monitoring during work may feel like performance tracking or disciplinary tool	Plain-language notice posted at the cell explaining what is collected and why; opt-out from analytics available without penalty or impact on cell access; Operator Advisory Group co-owns data governance decisions
Job displacement concern	Presence of the robot may signal future headcount reduction to operators	Pre/post survey tracks job security perception; management commitment to redeploy capacity gains to quality improvement formally documented; framing of robot as ergonomics tool, not productivity enforcer, embedded in all training materials

3.4 FSTP Support Service IV: Business support

Activity	Metric	KPI
ARISE website and partner social media updates (MMVR, UA, Pepri)	Posts published	≥1 per milestone (4 total)

Project presentations at industrial HRI and robotics conferences	Attended events	Minimum of 2
Peer-reviewed scientific publications	Submitted manuscripts	2 open-access publications by M12
Partner enablement tutorials and integration guides	Downloadable assets	≥1 tutorial video; ≥4 XR tutorials, 1 integration guide by M12

Exploitation/Business development plan.

For Pepri as an end user, the value is immediate and concrete. A single ergonomics cell adapted to their specific UR3e robot, their existing packaging station, and their toy rework SKUs eliminates the strain-intensive holding and rotation tasks that currently drive musculoskeletal risk on the line. The investment is bounded and the payback is measurable: with 3,000–4,500 productive hours per year and a documented reduction in near-misses, rework, and operator fatigue, the station generates a trackable ROI against a defined deployment cost. The short-term goal is validating this on one station and one product family. Mid-term, Pepri Toys can replicate the approach to adjacent stations using the same methodology, with significantly lower setup cost on each subsequent cell since the ErgoAssist skill and FIWARE models are already proven on their hardware.

For MMVR and UA as technology providers, the business reality is that POSTURE is a validated reference implementation. What is built here is specific to a UR cobot, a specific fixture design, and Pepri Toys' SKUs and production context. That specificity is actually the asset. Other factories face the same ergonomic challenges but have different robots, different stations, different operators, and different process constraints. What MMVR sells ist the proven capability to design, implement, and validate an ergonomics-first HRI cell tailored to a new client's context.

The commercialisation model is therefore case-study-driven. The Pepri Toys deployment becomes the anchor reference that MMVR presents to prospective clients , a real factory, real operators, real before-and-after ergonomics data. Each new engagement begins with a scoping and adaptation phase (understanding the client's robot, station, SKUs, and operator population), followed by an implementation phase that reuses the ErgoAssist architecture and FIWARE monitoring layer as a starting point but adapts them to the new environment. Revenue per engagement comes from the adaptation and integration work (estimated €30,000–€60,000 per station depending on complexity), fixture design or modification support, and optional managed analytics and support on a recurring basis (€5,000–€10,000 per station per year). Reuse of core components grows with each deployment, progressively improving margins as less bespoke development is needed.

The three-year trajectory is therefore realistic and grounded: Year 1 is Pepri Toys plus one or two early adopters identified through the demonstration and conference dissemination

activities, generating €0.3–0.6M primarily from services. Year 2 involves three to five new clients attracted by the published case studies and the ErgoAssist open reference, generating €0.8–1.4M as adaptation costs per site decrease. Year 3 expands to eight to twelve clients with a network of certified integrator partners handling straightforward replications, generating €1.5–2.5M with a growing share of recurring support revenue. The open ErgoAssist catalog submission and FIWARE templates serve primarily as lead generation and credibility tools for this pipeline, not as standalone revenue sources.

4. Experiment Budget

RESOURCES	Pepri	Metamedicsvr	University of Alicante	Cost justification
Personnel cost	37,392.00 €	63,595.00 €	42,108.00 €	Person months for each partner (explained at the proposal stage, unchanged)
Other Direct costs				
Travel costs for industry partner visits or Experiment-related presentations	300.00	€ 3,000.00 €	300.00 €	Three site visits to Alicante
Scientific publications			8,000.00 €	Two scientific publications.
Consumable	1,600.00	€		PPE, labels, and consumables for operator trials (€1,600). "

Equipment	1,500.00	€ Amount in €		Fixtures for the cell and HMI station peripherals, laptop, server etc.. "
Subcontracting costs		1,900.00 €		Costs to get featured for dissemination and commercialization
Overheads (25% of Personnel and Other direct costs)	10,198.00	€ 16,648.75 €	12,602.00 €	25%
Total estimated budget Third Party	50,990.00 €	85,143.75 €	63,010.00 €	(unchanged)
Requested Financial Support estimated	199,143.75 €			
Budget covered with own resources				